



Published in final edited form as:

Clin Cancer Res. 2021 August 01; 27(15): 4277–4286. doi:10.1158/1078-0432.CCR-21-0267.

OCaMIR – A noninvasive, diagnostic signature for early stage ovarian cancer: a multi-cohort retrospective and prospective study

Raju Kandimalla^{1,†}, Wei Wang^{2,3,†}, Fan Yu⁴, Nianxin Zhou⁵, Feng Gao⁶, Monique Spillman⁷, Lucie Moukova⁸, Ondrej Slaby^{9,10}, Bodour Salhia¹¹, Shengtao Zhou^{5,*}, Xin Wang^{2,12,*}, Ajay Goel^{1,13,*}

¹Center for Gastrointestinal Research; Center for Translational Genomics and Oncology, Baylor Scott & White Research Institute and Charles A. Sammons Cancer Center, Baylor University Medical Center, Dallas, Texas, USA.

²Department of Biomedical Sciences, City University of Hong Kong, Hong Kong, P.R. China.

³Department of Gynecology, Huzhou Maternity & Child Health Care Hospital, Huzhou, P.R. China.

⁴Department of Laboratory Medicine, West China Second Hospital, Sichuan University, Chengdu, P.R. China.

⁵Department of Obstetrics and Gynecology, Key Laboratory of Birth Defects and Related Diseases of Women and Children of MOE & State Key Laboratory of Biotherapy, West China Second University Hospital, Sichuan University and Collaborative Innovation Center, Chengdu, P.R. China.

⁶The Sixth Affiliated Hospital of Sun Yat-Sen University, Guangzhou, P.R. China.

⁷Department of Obstetrics and Gynecology, Baylor University Medical Center, Dallas, TX, USA.

⁸Masaryk Memorial Cancer Institute, Brno, Czech Republic.

⁹Central European Institute of Technology, Masaryk University, Brno, Czech Republic.

¹⁰Department of Biology, Faculty of Medicine, Masaryk University, Brno, Czech Republic.

*Correspondence to: Professor Ajay Goel, Department of Molecular Diagnostics and Experimental Therapeutics, Beckman Research Institute of City of Hope, 1218 S. Fifth Avenue, Suite 2226, CA 91016, USA, Tel./Fax: 626-218-4783, ajgoel@coh.org; Dr. Xin Wang, Department of Biomedical Sciences, City University of Hong Kong, 31 To Yuen Street, Hong Kong, P.R. China, Tel./Fax: +852 3442-2367, Xin.Wang@cityu.edu.hk; Professor Shengtao Zhou, Department of Gynecology and Obstetrics, West China Second Hospital, Sichuan University, Chengdu, 610041, P. R. China, taotaovip2005@163.com.

[†]Raju Kandimalla and Wang Wei share co-first authorship.

Authors' Contributions

Conception and design: A. Goel, X. Wang, S.-T. Zhou

Development of methodology: R. Kandimalla, W. Wei

Acquisition of data: F. Yu, N.-X. Zhou, F. Gao, M. Spillman, L. Moukova, O. Slaby, B. Salhia

Analysis and interpretation of data: R. Kandimalla, W. Wei

Writing, review, and/or revision of the manuscript: A. Goel, X. Wang, S.-T. Zhou, R. Kandimalla, W. Wei

Administrative, technical, or material support: R. Kandimalla, W. Wei, F. Yu, N.-X. Zhou, X. Wang, S.-T. Zhou

Study supervision: A. Goel, X. Wang, S.-T. Zhou

Disclosure of Potential Conflicts of Interest

No potential conflicts of interest were disclosed.

¹¹Department of Translational Genomics, University of Southern California, Los Angeles, CA, USA.

¹²Shenzhen Research Institute, City University of Hong Kong, Shenzhen, P.R. China.

¹³Department of Molecular Diagnostics and Experimental Therapeutics, Beckman Research Institute of City of Hope, Duarte, CA, USA

Abstract

Purpose: Due to the lack of effective screening approaches and early detection biomarkers, ovarian cancer (OC) has the highest mortality rates among gynecological cancers. Herein, we undertook a systematic biomarker discovery and validation approach to identify microRNA biomarkers for the early detection of OC.

Experimental Design: During the discovery phase, we performed small-RNA sequencing in stage-I high-grade serous OC ($n=31$), which was subsequently validated in multiple, independent datasets (TCGA, $n=543$; GSE65819, $n=87$). Subsequently, we performed multivariate logistic regression-based training in a serum dataset (GSE106817, $n=640$), followed by its independent validation in three retrospective datasets (GSE31568, $n=85$; GSE113486, $n=140$; Czech Republic cohort, $n=192$) and one prospective serum cohort ($n=95$). In addition, we evaluated the specificity of OCaMIR, by comparing its performance in several other cancers (GSE31568 cohort, $n=369$).

Results: The OCaMIR demonstrated a robust diagnostic accuracy in the stage-I high-grade serous OC patients in the discovery cohort (area under the curve [AUC] value=0.99), which was consistently reproducible in both stage-I (AUC=0.96) and all stage patients (AUC=0.89) in the TCGA cohort. Logistic regression-based training and validation of OCaMIR achieved AUC values of 0.89 (GSE106817), 0.85 (GSE31568), 0.86 (GSE113486), and 0.82 (Czech Republic cohort) in the retrospective serum validation cohorts, as well as prospective validation cohort (AUC=0.92). More importantly, OCaMIR demonstrated a significantly superior diagnostic performance compared to CA125 levels, even in stage-I patients, and was more cost effective: highlighting its potential role for screening and early detection of ovarian cancer.

Conclusions: Small RNA sequencing identified a robust non-invasive miRNA signature for early stage serous ovarian cancer detection.

Keywords

Ovarian cancer; miRNA; Diagnosis; Screening; Liquid biopsy

Introduction

Ovarian cancer (OC) is one of the most lethal gynecological malignancies worldwide (1). A primary reason for the morbidity and mortality associated with OC is that ~75% of women are diagnosed at advanced stages, at which the tumor has spread into the abdomen, resulting in poor 5-year survival rates (10–30%) (2). Although survival rates for early stage (stage I) OC can be as high as 90% (1), a lack of effective screening strategies and early detection markers prevents early diagnosis (3). Routine pelvic examination and transvaginal ultrasound are inefficient tools for population screening due to poor sensitivity (4). Although

serum protein level of the cancer antigen CA125 has been routinely used as a clinical OC marker for three decades, it couples poor sensitivity for detecting early stage cancers with limited specificity for OC (5,6). Two large randomized clinical trials PLCO (7) [Prostate, Lung, Colorectal, and Ovarian Cancer], and UKCTOCS (8) [United Kingdom Collaborative Trial of Ovarian Cancer Screening], tested CA125 with other markers and ultrasound failed to support use of CA125 as a screening tool, due to an absence of any overall survival benefit.

Previous attempts at discovering non-invasive biomarkers at the level of DNA methylation (9,10) or microRNA (miRNA) (11–14) have shown great promise, but no study has yet used stage I high-grade serous OC specimens to discover biomarkers for early diagnosis. Furthermore, previous studies lacked independent multicenter validations and no study has yet validated the identified markers in a prospective fashion. miRNAs are small, 18–24-nucleotide noncoding RNAs that play a pivotal role in gene regulation through post-transcriptional modification and are deregulated in OC (11–13,15,16) and other cancers (17–22). Importantly, miRNAs are regarded as more robust non-invasive biomarkers than messenger RNAs, as they are less prone to degradation in tissues, blood, stool, and other bodily fluids (23,24). However, the heterogeneity associated with sources of circulating, cell-free miRNAs appears to limit their specificity. To overcome these limitations and with the aim to identify robust miRNA biomarkers for early detection of OC, we undertook small RNA sequencing in primary tissue specimens from patients with stage I high grade serous OC. By employing robust statistical methods, we identified a panel of eight miRNAs (OCaMIR signature) that are highly accurate in identifying stage I high grade serous OC. In addition, we independently validated the identified tissue miRNA markers in multiple retrospective tissue and serum cohorts from various ethnicities by developing a multivariate logistic regression-based signature. We also evaluated the performance of the OCaMIR signature relative to the current gold standard CA125, as well as its cost effectiveness for screening purposes. We independently validated the 8-miRNA-based signature in a prospectively collected serum cohort, underscoring its future clinical translational potential for non-invasive detection and population screening of early stage OC.

Materials and Methods

Clinical samples and study design

A total of 661 tissue specimens and 1,152 serum specimens from 8 independent clinical cohorts were analyzed; corresponding patient demographic and clinicopathological characteristics are shown in Table 1. To establish a robust miRNA signature for non-invasive detection of early-stage OC, a genome-wide discovery approach followed by independent validation was employed in three phases (Supplementary Fig. S1).

For the initial tissue discovery phase, we acquired 22 fresh-frozen stage I high-grade serous epithelial OC specimens and 9 adjacent normal fallopian tube tissues from Asterand biobank collected at Baylor. Tumors were staged in accordance with FIGO (International Federation of Gynecology and Obstetrics) criteria and an expert pathologist reviewed all pathological data. All included tissue samples had >70% tumor cellularity. Given the sample size (22 OC and 9 normal tissues), our study was adequately powered (power > 0.95) to detect a

4-fold change ($\log_2$ -fold change > 2) at a false positive rate of 5%, with an average read depth of 1,903. For independent tissue validation, we used TCGA (25) and GSE65819 (26) publicly available cohorts, for which level-3 and preprocessed miRNA expression data as well as corresponding clinical data was downloaded from Broad GDAC Firehose (<http://firebrowse.org/>) and Gene Expression Omnibus (GEO, <https://www.ncbi.nlm.nih.gov/geo/>), respectively.

For the serum retrospective training and validation phase, we analyzed three previously published serum cohorts as well as one external retrospective validation serum cohort. The publicly available cohorts were GSE106817 (12), GSE113486 (27), and GSE31568 (28), for which the preprocessed miRNA expression values as well as clinical data was downloaded from GEO. The first retrospective serum cohort was collected at the Department of Comprehensive Cancer Care, Masaryk Memorial Cancer Institute (MMCI) Czech Republic, and sent to us on dry ice for RT-PCR-based validation. All samples were collected upon initial diagnosis of OC, before treatment. We received clinical and epidemiological data, as well as level of CA125 expression measured at MMCI. Healthy specimens from MMCI were collected from the screening population and confirmed to have no underlying disease or cancer.

For the serum prospective validation phase, we prospectively recruited patients ($n=44$) and healthy individuals ($n=51$) from Jan to June 2018 at the West China Hospital (WCH), Chengdu, China. The WCH sera were collected from newly diagnosed patients, before surgery with standard operating procedures. The exclusion criteria were: (a) history of preoperative treatment, (b) previous history of ovarian cancer, (c) other active malignancies confirmed. Expression levels of the 8 miRNAs were analyzed using RT-PCR at WCH. As a comparison, CA125 levels were measured using a Siemens ADVIA Centaur XP. A written informed consent was obtained from every participant, and the institutional review boards from all participating centers approved the study. This study adhered to Transparent Reporting of a multivariable prediction model for Individual Prognosis Or Diagnosis (TRIPOD) guidelines (29).

All procedures were conducted in accordance with the Helsinki Declaration. A written informed consent was obtained from all participants. This study was approved by the Institutional Review Boards of all participating institutions.

Tissue RNA isolation and small RNA sequencing

Total RNA was isolated from fresh-frozen OC tissues using AllPrep DNA/RNA/miRNA Universal Kit (Qiagen, Valencia, CA, USA). Briefly, next-generation sequencing library construction for miRNA from tissue was performed using a modified protocol for the Truseq Small RNA Kit (Illumina) with 200 ng total RNA input. The quality of individual libraries was assessed using a High Sensitivity DNA Kit (Agilent). Libraries were size selected individually (~148 nt) by gel electrophoresis using a Pippin HT instrument (Sage Science). Efficiency of size selection was assessed using a High Sensitivity DNA Kit. Libraries were equimolar-pooled; pooled libraries were quantitated via qPCR using a KAPA Library Quantification Kit, Universal (KAPA Biosystems) prior to sequencing on an Illumina HighSeq 2500 with single-end 35-base read lengths at an average of 10 million reads

per sample. For preprocessing, Illumina small RNA-seq 3' adapters were trimmed using *cutadapt* software (30). Post-trimming, all retained sequences contained high-quality scores and peaks concentrated at 22 nt, representing miRNAs. Preprocessed reads were aligned to a human reference genome (human genome build 38) and annotated using GENCODE miRNA annotation (Release 24). Sequence alignment and quantification of miRNA counts were performed using STAR software (31). Differential miRNA expression was analyzed using the *limma* (32) software package. The raw sequencing data is accessible in GEO database (accession number: GSE127873).

Serum miRNA isolation and RT-PCR for independent retrospective and prospective validation

RNA enriched for small non-coding RNAs was extracted from sera (200 μ L) using an miRNeasy serum kit (Qiagen). For normalization of sample-to-sample variation during RNA isolation procedures, synthetic *C. elegans* miRNA (cel-miR-39, Qiagen) was added to each denatured serum. miRNA expression was quantified using TaqMan miRNA real-time RT-qPCR assays (4427975) and QuantStudioTM 7 Flex Real-Time PCR System (Applied Biosystems, Foster City, CA). The following primers (assay ID) were used for the OCaMIR signature RT-qPCR assay: hsa-miR-202-3p (#2363), hsa-miR-205-5p (#509), hsa-miR-508-3p (#1052), hsa-miR-509-3-5p (#2155), hsa-miR-513c-5p (#2756), hsa-miR-183-5p (#2269), hsa-miR-182-5p (#2334), and hsa-miR-513b-5p (#2757) (Thermo Fisher Scientific, Waltham, MA). Several endogenous controls including U6 (#1973), RNU44, hsa-miR-16-5p, and hsa-miR-103a-3p (Thermo Fisher Scientific) were tested for data normalization. All the endogenous controls were expressed without any difference between healthy controls and cancer specimens. We used U6 for data normalization of the Czech and West China cohorts. Expression of each miRNA was calculated using the $2^{-\Delta\Delta CT}$ method.

Statistical analysis

Statistical analyses were performed using R 3.5.2 (<https://cran.r-project.org/>), IBM SPSS version 23 (<https://www.ibm.com/analytics/spss-statistics-software>), and GraphPad Prism version 6.0 (<https://www.graphpad.com/scientific-software/prism/>). All statistical tests were two-sided and Benjamini–Hochberg (BH) method was used to correct for multiple hypothesis testing wherever applicable. Power calculation for the tissue discovery phase was performed using the *RNASeqPower* package (33) in R. Differential miRNA expression analysis was performed between cancer and normal tissues from the small RNA sequencing data using *limma* (32). Differential miRNAs were identified based on 1) $|\log_2$ fold change| >1 , and 2) $P < 0.05$, and candidate miRNAs were prioritized by 1) $|\log_2$ fold change| >2 , and 2) BH-adjusted $P < 0.05$. The Principal components analysis was performed by ‘stats’ package in R. Eight miRNAs significantly upregulated in OC tissues were employed to build a multivariate logistic regression model from the discovery cohort, and the scoring formula derived from the model was applied to TCGA and GSE65819 for independent tissue validation. As expected, the dynamic expression range of the miRNAs between tissue and serum varied; therefore, we trained a serum-specific logistic regression model using the expression levels of the 8 miRNAs in the GSE106817 dataset. The scoring formula derived from this serum training cohort was applied to the publicly available datasets

(GSE31568 and GSE113486), the Czech Republic retrospective serum validation cohort, and the prospective West China serum validation cohort. The serum-specific scoring formula derived from the trained multivariate logistic regression model was: $\text{logit}(P) = -1.3374 + (0.0887 \cdot \text{MIR202}) + (0.1622 \cdot \text{MIR508}) + (0.3776 \cdot \text{MIR509-3}) + (-0.0381 \cdot \text{MIR205}) + (0.1101 \cdot \text{MIR182}) + (0.1038 \cdot \text{MIR513C}) + (-0.0690 \cdot \text{MIR183}) + (0.2317 \cdot \text{MIR513B})$. To determine the accuracy of the OCaMIR signature, the predictive risk probabilities calculated from the formula were used to plot receiver operating characteristic (ROC) curves with 95% confidence intervals. All ROC curves presented in our results were shown with 95% CI. CI was calculated by 2,000 bootstrap replicates. The optimal cut-off points for ROC curves were determined using Youden's index in the 'pROC' package. Based on the ROC Curves, we calculated the sensitivity, specificity, positive predictive value (PPV), and negative predictive value (NPV) of OCaMIR across all the cohorts using the reportROC package (34). CA125 was dichotomized at 35 u/ML in the Czech and West China cohorts. Direct comparison of OCaMIR with CA125 was performed using ROC curves.

MiRNA regulatory network inference and functional analysis

A miRNA-mRNA network was constructed to study the regulatory functions of the candidate miRNAs. More specifically, for each of the miRNAs we identified its target mRNAs based on the following criteria: 1) the miRNA-mRNA interactions have been experimentally validated based on the miRTarBase database (V8) (35); 2) the mRNAs are differentially expressed between tumor and normal samples ($|\log_2 \text{fold change}| > 1$ & BH-adjusted $P < 0.05$) in the TCGA dataset. Functional analysis was performed based on hypergeometric tests using 'clusterProfiler' package (36), with KEGG pathways (37) and Hallmark gene sets retrieved from the MSigDB Database (v7.0) (38). P-values were corrected for multiple hypothesis testing using the Benjamini-Hochberg (BH) Procedure, and BH-adjusted p-value < 0.05 was considered statistically significant.

Cost-effectiveness analysis

Cost-effectiveness analysis (CEA) for comparing the overall costs and health benefits of conducting large-scale screening using the OCaMIR signature vs. current clinical practice without screening was performed for a hypothetical cohort through Markov modeling. Markov modeling of CEA was performed using the R package heemod (39). The details of clinical assumptions of the hypothetical cohort can be found in appendix (Supplementary Table S5 and S6).

Results

Discovery of an OCaMIR signature from genome-wide small RNA sequencing data

To establish a robust miRNA signature for non-invasive detection of early-stage serous OC, we designed our study to encompass three stages: 1) tissue discovery and independent validation, 2) serum retrospective training and validation, and 3) serum prospective validation (Supplementary Fig. S1 and Table 1). In the discovery phase, we used small RNA sequencing to identify 143 miRNAs that were differentially expressed ($P < 0.05$ and $|\log_2 \text{fold change}| > 1$) between stage I serous OC and normal tissues (Fig. 1A). Of these, we prioritized eight miRNAs (BH-adjusted $P < 0.05$ and $\log_2 \text{fold change} > 2$) that

are upregulated —MIR182, MIR183, MIR202, MIR205, MIR508, MIR509–3, MIR513B, and MIR513C (Fig. 1B; OCaMIR signature). To investigate whether these candidate miRNAs are functionally related to OC development, we constructed a miRNA-mRNA regulatory network based on experimentally validated miRNA-target interactions obtained from miRTarBase database (V8). In total, 148 differentially expressed genes ($|\log_2$ fold change| > 1 and BH-adjusted $P < 0.05$) between TCGA tumor and normal samples were identified to be the targets of these miRNAs (Supplementary Fig. S2; Supplementary Table S1). As expected, these miRNA target genes are significantly enriched in cancer-related signaling pathways and cancer hallmark gene sets such as pathways in cancer, focal adhesion and epithelial to mesenchymal transition (Supplementary Fig. S2; Supplementary Table S2). To evaluate the diagnostic accuracy in stage I OC, we developed a logistic regression model and derived risk probabilities from the linear combination of the eight miRNAs (Fig. 1C). In the tissue discovery cohort, OCaMIR achieved an area under the curve (AUC) of 0.99 for detecting stage I OC (Fig. 1C). To evaluate the robustness of OCaMIR, we applied the risk-scoring formula derived from the logistic regression model to two independent, publicly available datasets (TCGA and GSE65819 cohorts). Principal component analysis showed that the eight miRNAs can discriminate tumor and normal group samples (Fig. 1D). Consistent with discovery cohort results, OCaMIR achieved an AUC of 0.96 in detecting stage I OC in the TCGA cohort, and an AUC of 0.89 and 0.83 for detecting all stages of OC in the TCGA and GSE65819 cohorts, respectively (Fig. 1E). These results confirm the proficiency of OCaMIR for OC detection at the tissue level.

Independent retrospective training and validation of OCaMIR signature in sera across international multicenter cohorts

To evaluate the potential use of the OCaMIR signature as a non-invasive diagnostic test, we analyzed three publicly available serum miRNA expression cohorts (GSE106817, GSE31568, GSE113486) and one external retrospective serum validation cohort (RT-PCR to analyze miRNA from Czech Republic cohort). As expected, the eight individual miRNAs were upregulated in OC compared to expression in healthy specimens across all the serum validation cohorts (Table 2; Supplementary Fig. S3).

To tailor OCaMIR for non-invasive detection in sera, we used the GSE106817 miRNA microarray dataset (320 OC vs. 320 healthy sera) to retrain a serum-specific logistic regression model with all eight miRNAs. The risk-probability plots (OCaMIR risk probability) differentiate between healthy and OC sera (Fig. 2A). OCaMIR achieved an AUC of 0.89 for detecting OC (Fig. 2A) in the GSE106817 serum training cohort. Furthermore, OCaMIR obtained a sensitivity of 88%, specificity of 74%, PPV of 77%, and NPV of 86% in the GSE106817 serum training cohort (Supplementary Fig. S4 and Supplementary Table S3).

We subsequently applied the OCaMIR logistic regression model developed from the GSE106817 serum training cohort to three independent retrospective serum cohorts: 1) GSE31568 miRNA microarray dataset (OC, $n=15$; healthy, $n=70$), 2) GSE113486 miRNA microarray dataset (OC, $n=40$; healthy, $n=100$), and 3) Czech Republic RT-PCR-based expression dataset (OC, $n=96$; healthy, $n=96$). The OCaMIR risk-probability plots

differentiated between healthy and OC sera among all the serum validation cohorts (Fig. 2, B to D). OCaMIR achieved an AUC of 0.85 (GSE31568), 0.86 (GSE113486), and 0.82 (Czech Republic) for the detection of OC including all stages (Fig. 2, B to D, respectively). OCaMIR sensitivity, specificity, PPV, and NPV for each serum validation cohort are presented in Supplementary Fig. S4 and Supplementary Table S3. These results highlight the applicability of our eight-miRNA signature for non-invasive detection of early OC in serum.

OCaMIR signature is highly accurate at detecting stage I OC in serum, superior to CA125, and highly specific for OC detection

Two of our retrospective serum cohorts included stage I OC. Our stage-wise analysis revealed that, in line with tissue-based results, OCaMIR was highly accurate at detecting stage I OC, with an AUC of 0.90 and 0.86 in the GSE106817 and Czech Republic serum cohorts, respectively (Fig. 3, A and B). OCaMIR sensitivity, specificity, PPV, and NPV for stage I OC detection are presented in Supplementary Table S4. The Czech Republic cohort dataset includes CA125 levels for both healthy and OC specimens, which allowed us to compare the performance of OCaMIR to CA125 in OC detection. As shown above, OCaMIR achieved an AUC of 0.82 for all stages (Fig. 2D) and 0.86 for stage I (Fig. 3B) OC. In contrast, CA125 performance was significantly inferior, with an AUC of 0.80 for the detection of all stages of OC ($P=0.03$, ROC test) and an AUC of 0.73 for stage I OC ($P=0.09$, ROC test) (Fig. 3B).

To evaluate the specificity of OCaMIR for OC detection, we used the GSE31568 cohort, for which miRNA expression data was available for various diseases including OC. OCaMIR achieved an AUC of 0.85 in OC, whereas detection accuracy for other diseases (acute myocardial infarction, lung cancer, melanoma, multiple sclerosis, pancreatic cancer, pancreatitis, prostate cancer, sarcoidosis, stomach cancer, and Wilms tumor) ranged from 0.19 to 0.71 (Fig. 3D). This confirms that the OCaMIR signature is superior to the conventional marker CA125 and is highly specific for detection of OC.

OCaMIR signature is accurate in prospective serum validation and is an economical option for non-invasive population screening

To validate the ability of OCaMIR to detect OC prospectively, we independently validated it in prospective international samples collected at WCH. We used RT-PCR to profile the eight OCaMIR signature miRNAs at WCH using the standard operating procedure established during our retrospective validation. We applied the logistic regression model developed during the retrospective phase to the prospective setting and generated risk probabilities (Fig. 2E). Consistent with previous results, OCaMIR accurately detected patients with all stages of OC, with an AUC of 0.92, sensitivity of 82%, specificity of 86%, PPV of 84%, and NPV of 85% in the prospective cohort (Supplementary Table S3). The AUC for stage I OC in the prospective cohort was 0.81 (Fig. 3C). Consistent with the Czech Republic retrospective serum validation data, OCaMIR performed better than CA125 for the detection of stage I and all stage OC in the WCH prospective cohort (Fig. 3C).

Using the sensitivity and specificity achieved in the prospective validation cohort, we performed CEA to evaluate the cost-benefit of using our miRNA signature for large-scale screening within the clinical environment of mainland China. The targeted high-risk population is 55–80-year-old Chinese women. We used a Markov model-based CEA to estimate that a large-scale OCaMIR screening would be cost-effective relative to the current practice of no screening [ICER=CNY 5139.4/QALY] (ICER, incremental cost-effectiveness ratio; CNY, Chinese Yuan Renminbi; QALY, quality-adjusted life year; Supplementary Table S5 and S6). These results highlight the validity of the OCaMIR signature across distinct laboratories and its applicability in the prospective setting, making it a viable option for high-risk population-based screening.

Discussion

Although advances in surgical procedures and chemotherapy regimens have moderately improved OC survival rates, most women are diagnosed at very late stages, when a cure is unlikely (40). Thus, there is a dire need to identify clinically translatable, highly accurate, non-invasive, and cost-effective biomarkers that aid in early detection and population screening of high-risk populations, which may also improve OC survival outcomes.

Owing to the stability of miRNAs in serum and plasma, they show promise for identification of non-invasive early detection biomarkers in various cancers (13,18,41) including OC (11–13,16,42–47). Although several studies identified miRNAs in OC, limitations remain, which we overcame as follows: 1) To identify early detection markers, it is critical to profile stage I OC during the discovery phase; ours was the first study to perform small RNA sequencing in early stage OC tissues. 2) We independently validated the identified tissue-based markers in ethnically diverse TCGA and GSE tissue cohorts, providing evidence of their utility across diverse patient populations. 3) We applied the tissue-derived markers to multiple independent, ethnically diverse serum-based OC cohorts, which further emphasized OC specificity; consistent with tissue data, OCaMIR was effective across diverse patient populations in sera. 4) Our retrospective serum validation cohort consisted of several methodological platforms, including microarray-, sequencing-, and qRT-PCR-based methods, which suggests that the miRNA markers we identified are consistent across different platforms. 5) Ours was the first study to evaluate the identified markers in a prospective setting by implementing a standard operating procedure in an independent lab, suggesting that the OCaMIR signature can be easily adapted into routine clinical labs, following further investigation. 6) We evaluated OCaMIR specificity for OC relative to various other cancers and diseases. 7) We showed OCaMIR performed superior to the current gold standard CA125 in both retrospective and prospective settings.

One of the limitations of our study is the lack a large number of stage I high-grade serous ovarian cancer in the discovery phase. However, we have showed that the sample size in the discovery phase have enough statistical power. Moreover, we were able to validate the markers across various independent tissue and serum cohorts, which further warrants the translational potential of OCaMIR upon further large-scale randomized prospective studies.

Although high-risk population screening for OC may improve the outcomes of this deadly disease, previous large-scale OC prospective screening studies aimed at postmenopausal, average-risk women (e.g., PLCO (7), UKCTOCS (8)) failed to provide any statistically significant improvement in OC mortality associated with screening. These studies investigated the screening potential of CA125 and/or transvaginal ultrasound and compared the outcomes to no-screening populations. In these trials, the cumulative false-positive rates ranged from 9.8% to 44% (48), which can lead to unnecessary surgical procedures and complications. Therefore, the OCaMIR signature we developed was effective across various patient populations and sample sources, which is very exciting and holds great promise for future investigations.

Of the eight miRNAs we identified, MIR205 (49–51), MIR202 (52,53), MIR182, and MIR183 (54) were reported previously as diagnostic markers in OC. Although the functional relevance of diagnostic markers is of limited concern as long as diagnostic accuracy is high, most of the OCaMIR miRNAs were previously reported to be associated with OC progression via multiple pathways. For example, several studies showed that MIR205 is involved in OC progression via suppression of *SMAD4*, *PTEN*, and *ZEB1* tumor suppressor genes (55–58). Recently Steven et al. showed that MIR205, MIR182, and MIR183 expression is highly elevated in omental metastases and chemoresistant OC (59). MIR182 was reported to promote cell growth, invasion, and chemoresistance by targeting programmed cell death 4 (60); was suggested to be an attractive in vivo therapeutic target in OC (61); and is a negative BRCA1 regulator that is significantly overexpressed in high-grade serous OC (62). We recently identified MIR508 as associated with mesenchymal subtype OC and to drive poor prognosis (63). Taken together, these studies attest to the significance of the eight-miRNA OCaMIR signature that we identified.

Supplementary Material

Refer to Web version on PubMed Central for supplementary material.

Funding:

This work was supported by CA72851, CA181572, CA184792, and CA187956 grants from the National Cancer Institute, US National Institutes of Health to Ajay Goel. In addition, this project was supported by a pilot project funded by the Baylor Scott & White Research Institute and Translational Genomics Institute partnership funding to Ajay Goel and Bodour Salhia. Support also included grants from the Research Grants Council of the Hong Kong Special Administrative Region, China (Project No. CityU 11102317, 11103718, 11103619, R4017–18, C4041–17GF), a grant from Guangdong Basic and Applied Basic Research Foundation (Project No. 2019B030302012), and a grant from the National Science Foundation for Young Scientists of China (Project No. 81802384), awarded to Xin Wang. Support also included grants from the National Natural Science Foundation of China (81822034, 81773119, and 81402396), National Key Research and Development Program of China (2017YFA0106800, 2018YFA0109200), Sichuan Science-Technology International Cooperation Project (2019YFH0144), and Direct Scientific Research Grants from West China Second Hospital, Sichuan University (KS021) awarded to Shengtao Zhou.

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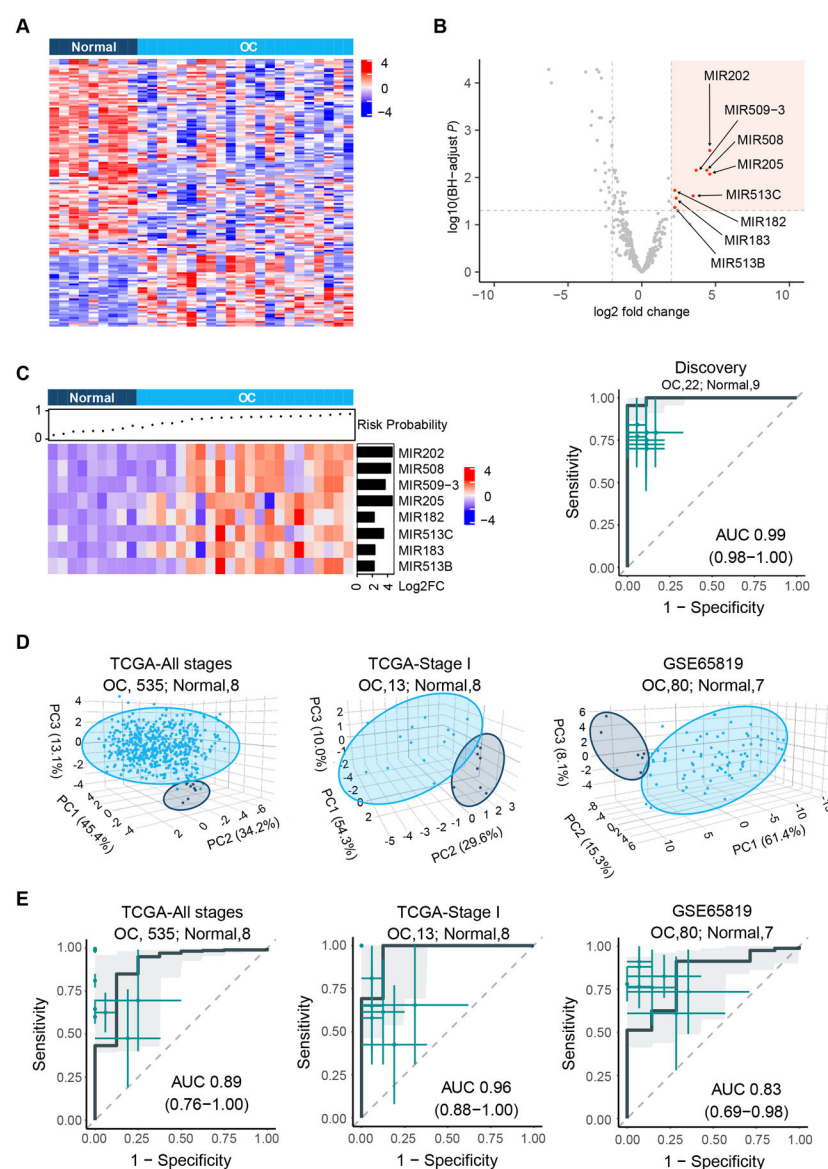
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Translational Relevance:

Due to the lack of effective screening approaches and early detection biomarkers, ovarian cancer (OC) has the highest mortality rates among gynecological cancers. Previous attempts at discovering non-invasive miRNA-based biomarkers have shown great promise, but no study has yet validated the markers in ethnically diverse retrospective and prospective cohorts. Here, we performed a systematic and comprehensive biomarker discovery using stage I high-grade serous OC specimens, we independently validated the identified tissue-based markers, providing evidence of their utility across diverse patient populations. We applied the tissue-derived markers to multiple independent, ethnically diverse serum-based OC cohorts, which further emphasized OC specificity; consistent with tissue data, OCaMIR was effective across diverse patient populations in sera. OCaMIR demonstrated a significantly superior diagnostic performance compared to CA125 levels, even in stage-I patients, and was significantly more cost effective; highlighting its potential role for screening and early detection of ovarian cancer.

**Figure 1.**

Discovery and independent tissue validation of OCaMIR signature. **A**, Heatmap illustrating differentially expressed miRNAs between the stage I serous ovarian cancer (OC) and tumor adjacent normal tissues in the discovery cohort ($|\log_2 \text{fold change}| > 1$ and $P < 0.05$). **B**, Volcano plot highlighting the eight candidate miRNAs prioritized based on the comparison between stage I OC and normal tissues ($\log_2 \text{fold change} > 2$, BH-adjusted $P < 0.05$). **C**, Heatmap of the eight candidate miRNAs' expression and ROC curve for the discovery dataset. Risk probabilities derived from the 8-miRNA signature. Heatmap of expression (left) and a ROC curve (right) are shown for the 8 candidate miRNAs in the discover cohort. In the heatmap, predictive probabilities were calculated using the logistic regression model, shown in the top panel; expression fold changes are shown in the right panel. **D**, Principle component analysis shows the eight miRNAs can discriminate OC and normal samples in TCGA (all stages), TCGA (stage I) and GSE65819 (all stages) cohorts. **E**, ROC curves

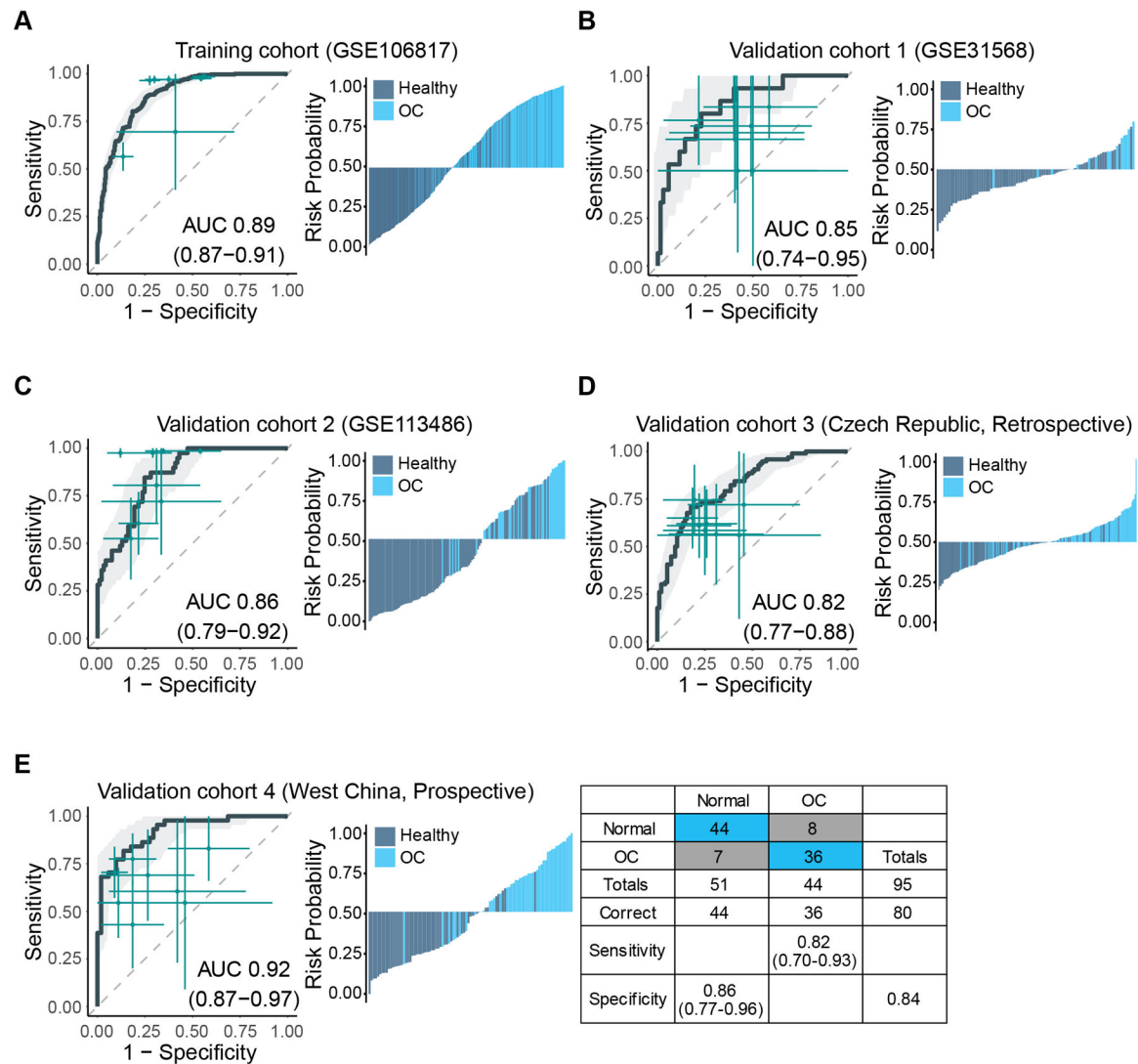
showing independent tissue validation of the 8-miRNA signature in TCGA (all stages), TCGA (stage I) and GSE65819 (all stages) cohorts. ROC curves are shown with 95% CI.

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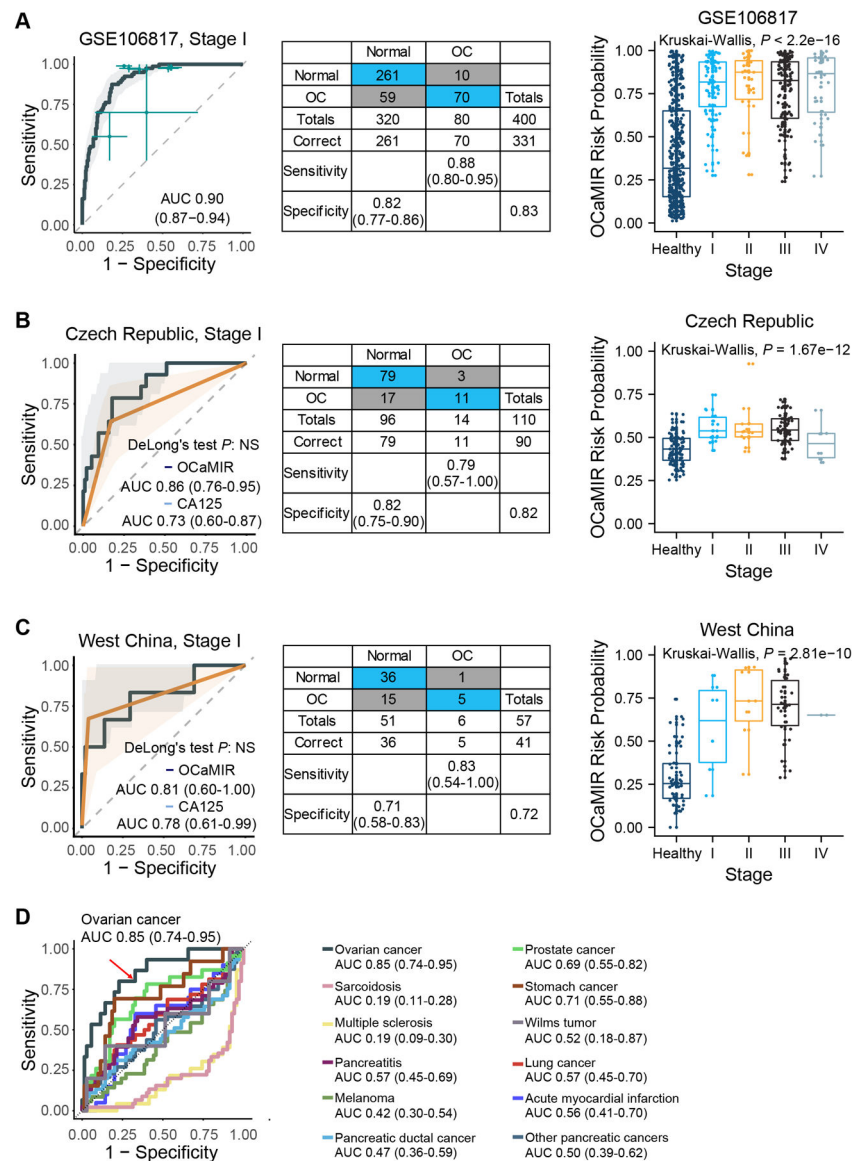
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**Figure 2.**

OCaMIR signature serum independent training and validation across four retrospective and one prospective cohort. ROC curves and risk-probability plots of OCaMIR: **A**, Serum retrospective training cohort; GSE106817 miRNA microarray. **B**, Serum retrospective validation cohort 1; GSE31568 miRNA microarray. **C**, Serum retrospective validation cohort 2; GSE113486 miRNA microarray. **D**, Serum retrospective validation cohort 3; Czech Republic RT-PCR. **E**, Serum prospective validation cohort; West China RT-PCR. The predictive performance of OCaMIR in the West China cohort is summarized in the confusion matrix.

**Figure 3.**

Evaluation of OCaMIR signature performance in detecting stage I ovarian cancer (OC) in serum and specificity for OC detection. ROC curves, confusion matrices and stage-wise risk-probability plots of OCaMIR: **A**, Serum retrospective training cohort; GSE106817 miRNA microarray, **B**, Serum retrospective validation cohort 3; Czech Republic RT-PCR, **C**, Serum prospective validation cohort; West China RT-PCR and **D**, ROC curves of OCaMIR assay across various diseases including OC in serum retrospective validation cohort 1; GSE31568 miRNA microarray.

Table 1.

Clinical characteristics of patients and healthy participants in the tissue and serum cohorts

	Tissue Cohort		Serum Cohorts (retrospective&prospective)			
	Training Cohort	Validation Cohort 1	Validation Cohort 2	Training Cohort	Validation Cohort 1	Validation Cohort 2
OC	Baylor; n = 31	TCGA; n = 543	GSE65819; n = 87	GSE106817; retrospective; n = 640	GSE31568; retrospective; n = 85	GSE113486; retrospective; n = 140
Age, median (range), y	22 (71%) 59 (46–78)	535 (99%) 59 (26–89)	80 (92%)	320 (50%)	15 (18%)	40 (29%) 54 (31–81)
Histopathology						
HGSOC	22 (100%)	535 (100%)	80 (100%)	320 (100%)	NA	NA
Cancer stage						
I	22 (100%)	13 (2%)		81 (25%)		14 (15%)
II		24 (4%)		30 (9%)		13 (14%)
III		411 (77%)		112 (35%)		60 (62%)
IV		81 (15%)		32 (10%)		6 (6%)
Unstaged		14 (3%)		66 (21%)		3 (3%)
Serum CA125, ng/ml						1109.1±204.8
Healthy participants	9 (29%) 44 (31–52)	8 (1%)	7 (8%)	320 (50%)	70 (82%)	100 (71%)
Age, mean (range), y						53 (42–79)
Serum CA125, ng/ml						0.16±0.37

HGSOC, high grade serous ovarian cancer

Table 2.

Training phase													
Training Cohort (GSE106817)			Validation Cohort 1 (GSE31568)		Validation Cohort 2 (GSE113486)		Validation Cohort 3 (Czech Republic)		Validation Cohort 4(West China)		Logistic regression model		
miRNA	OC vs. Healthy	AUC	OC vs. Healthy	AUC	OC vs. Healthy	AUC	OC vs. Healthy	AUC	OC vs. Healthy	AUC	Estimate	z value	Pz(> z)
MIR202	up	0.66	ns	0.42	up	0.68	up	0.70	up	0.74	0.09	2.01	0.04
MIR508	up	0.78	ns	0.52	up	0.72	up	0.73	up	0.83	0.16	3.07	< 2.00E-16
MIR509-3	up	0.80	ns	0.65	up	0.77	up	0.80	ns	0.57	0.38	7.70	1.35E-14
MIR205	up	0.66	ns	0.37	up	0.65	up	0.66	ns	0.49	-0.04	-0.76	0.44
MIR182	up	0.70	up	0.86	up	0.70	up	0.71	ns	0.43	0.11	2.41	0.02
MIR513c	up	0.64	up	0.71	up	0.91	up	0.96	up	0.73	0.10	1.90	0.06
MIR183	up	0.67	up	0.67	ns	0.40	ns	0.39	ns	0.50	-0.07	-1.47	0.14
MIR513b	up	0.81	ns	0.52	up	0.77	up	0.75	up	0.80	0.23	4.43	9.23E-06
(Intercept)											-1.33	-8.51	< 2.00E-16